A Computational Framework to Uncover Gray Areas in Tax Legislation

February 7, 2026

1 Abstract

It is often said that every law has its loophole, meaning that legal rules frequently contain technical flaws, ambiguities, or unspecified details that can be legally exploited to bypass the intent of the law. In the context of taxation, such gray areas contribute to regressivity, unfairness, and substantial revenue losses, costing governments billions of dollars worldwide. reducing tax disputes increasingly requires more efficient and cost-effective strategies than traditional ex post resolution. In this context, natural language processing models offer a scalable and adaptable opportunity to anticipate gray areas directly from legal texts. However, systematic approaches that leverage these tools to identify potential sources of tax norm abuse at scale remain limited [2]. This paper asks whether language-based computational methods can contribute to the systematic identification of gray areas in tax legislation. To address this question, we propose a methodology to define, identify, and annotate legal provisions according to their potential gray-zone risk. We apply this methodology to Colombian Free Trade Zone (FTZ) legislation and construct, to our knowledge, the first systematic dataset of tax norms annotated for gray-area-related non-compliance risk. Using this dataset, we implement a classification task based on TF–IDF representations and compare its performance with transformer-based models, as well as with models trained on non-annotated corpus. The classification model achieves around 70% precision in its best specification

Keywords: Legal NLP, Computational Legal Analysis, Tax Law Analysis, Legal Text Classification

2 Introduction

Every law has its loophole. Colombian Free Trade Zone legislation offers a clear illustration of how legal uncertainty does not disappear with reform, but rather mutates over time. In 1958, Free Trade Zones were introduced as a policy instrument to stimulate investment and employment through the production of manufactured goods in Colombia. Law 105 of 1958 granted tax benefits to firms

established in delimited coastal areas, under the expectation that these firms would engage in productive activity [18]. The legal design, however, allowed domestic firms to locate in these zones, access preferential treatment, and import goods without carrying out any real production. Subsequent reforms attempted to close this gap. By the mid-1980s, legislation distinguished between commercial and industrial zones, granting tax benefits only to firms that "develop the industrialization process of a product."[18] Yet the notion of industrialization remained undefined. In practice, firms carried out purely formal transformations, such as labeling or repackaging, to qualify for the benefit. More recent reforms followed the same pattern. Law 2277 of 2022 replaced production requirements with the obligation to submit an export plan reviewed by a public official [19]. Even under this framework, approximately 75 percent of firms operating in Free Trade Zones continue to produce for the domestic market [8]. Each reform closed a visible loophole and opened a new one, rooted in legal ambiguity.

Gray areas of this kind are not marginal defects. They are structural features of tax law. Ambiguity, incompleteness, discretionary language, and differential treatment generate spaces where legal interpretation determines tax outcomes. In taxation, this problem carries particular weight [9]. Through these gaps, firms pay less tax, public revenue declines, and fiscal capacity erodes. The global scale of the problem is substantial. Tax controversy imposes high costs on both taxpayers and administrations. In 2022, firms involved in transfer pricing disputes reported average expenditures of USD 56.3 million related to assessments, USD 24.7 million in penalties and interest, and USD 21.3 million in legal fees [10]). All 825 surveyed businesses reported at least one recent tax dispute, and more than 70 percent indicated that tax audits have increased in recent years as governments seek to close loopholes and curb avoidance [25]. These figures reflect a system that reacts after exploitation occurs, rather than preventing it at the drafting stage.

Despite its relevance, the problem remains weakly systematized in the literature. Governments and tax authorities address gray areas on a case-by-case basis, often through audits, litigation, or ad hoc administrative guidance. Computational approaches that attempt to anticipate gray areas directly from statutory language remain in an early stage. While recent studies explore how language models interact with tax law, there is still no systematic framework that links linguistic features of legal texts to the risk of norm abuse at scale.

This paper addresses the following research questions: How can gray areas in tax legislation be defined in a way that is both legally meaningful and computationally tractable? Can linguistic features of statutory texts signal provisions that are prone to exploitative interpretation? To what extent can language-based models recover expert assessments of gray-area risk, and where do they fail? What is the relation between tax law complexity and the gray area risk detection?

The paper contributes to the literature in three ways. First, it proposes an operational framework to identify gray-area triggers grounded in legal theory and expert practice. Second, it constructs the first systematic dataset of

Colombian tax provisions annotated for gray-area-related risk, using Free Trade Zone legislation as a case study. Third, it evaluates the ability of standard text representations and legal language models to detect these triggers, and relates their performance to legal text complexity.

The remainder of the paper is structured as follows. Section 2 reviews related work on ambiguity detection, tax uncertainty, and computational approaches to norm abuse. Section 3 presents the annotation framework, dataset construction, and methodology. Section 4 reports the empirical results. Section 5 discusses the implications and limitations of the findings. Section 6 concludes.

3 Related Work

Detecting exploitable ambiguities in legal texts requires understanding how language creates interpretive flexibility that enables norm abuse. We organize prior research into three areas: natural language processing approaches to ambiguity, computational methods targeting tax uncertainty, and alternative techniques using genetic algorithms.

3.1 NLP and Ambiguity Detection

The challenge of disambiguating meaning in text has long been recognized as difficult. Navigli [20] shows that fine-grained disambiguation achieves only 65–70% accuracy, establishing fundamental limits for semantic tasks. Bhattacharya et al. [1] probe BERT and GPT-2 representations and find that ambiguity is poorly encoded, with performance barely exceeding TF–IDF baselines. They further show that middle layers outperform upper layers, suggesting information loss at higher abstraction levels. Wildenburg et al. [24] demonstrate that models recognize underspecification above chance but fail to recognize when underspecified sentences license continuations that their specified counterparts do not.

In legal and technical domains, the picture becomes more complex. Guitton et al. [15] develop annotation protocols for open-texture in EU legal texts, achieving inter-annotator agreement of $\kappa = 0.70$ after reconciliation, up from 0.24 initially, revealing both the difficulty and tractability of annotating abstract legal concepts. Ferrari and Esuli [12] use domain-specific word2vec models to identify ambiguous terms across Wikipedia corpora, achieving Kendall's τ up to 0.88 for two-domain comparisons, though non-expert annotators miss domain-specific nuances. Ezzini et al. [11] achieve approximately 80% precision detecting ambiguity in requirements engineering, with domain-specific corpora improving detection by approximately 33%. Kim et al. [17] develop task-specific embeddings for smart speaker command ambiguity, finding that they outperform off-the-shelf embeddings. Taken together, these studies show that domain-specific representations consistently outperform generic ones, that annotation of abstract properties remains challenging, and that distinguishing whether models

capture genuine semantic properties or merely surface lexical patterns remains an open question.

3.2 Computational Approaches to Tax Uncertainty

Recent studies show that language models exhibit non-trivial capabilities in analyzing legal texts, while systematically diverging from expert judgment. This divergence suggests that such models are better suited to augment, rather than replace, expert-driven approaches. Blair-Stanek et al. [2] show that OpenAI's `o1` model can generate novel tax strategies. However, their subsequent evaluation [3] reveals that Spearman's ρ for whole-strategy grading ranges only between 0.30 and 0.48 when compared to expert grades, far below the 0.89 achieved for simpler tasks. Guitton et al. [14] find that `gpt-3.5-turbo` detects open-texture in GDPR text with $F_1 = 0.84$, yet only 42% of the terms flagged by humans appear in model output, and inter-annotator agreement among LLMs remains low ($\kappa = 0.01$–0.36). Fratrič et al. [13] combine large language models with Prolog-based planning to expose loopholes in multinational structures, though the formalization of legal rules by LLMs remains insufficient without human fine-tuning.

3.3 Other Approaches: Genetic Algorithms

Hemberg et al. [16] employ co-evolutionary genetic algorithms modeling adversarial taxpayer-auditor dynamics, where tax schemes and audit detection rules evolve simultaneously. This structural approach operates on transactional representations rather than statutory text, but it is formalized over the idea that "each time the IRS changes the tax code the tax evaders react by finding new ambiguities."

4 Methodology

A folk saying states that every rule has its loophole. In tax law, the exploitation of such loopholes entails substantial economic costs, amounting to billions in lost public revenue. Addressing this challenge requires scalable approaches capable of anticipating the emergence of these gray areas before they are systematically exploited, both to prevent tax avoidance and to provide clearer rules for taxpayers. To this end, we employ natural language processing to identify provisions prone to ambiguity and strategic interpretation before systematic exploitation occurs. We define *gray area triggers*—linguistic and structural features signaling interpretive risk—and use expert annotations to train and compare classification models. We then evaluate model robustness, examine information encoded in annotations through embeddings, and assess how predictive performance relates to legal text complexity.

4.1 Framework for Gray-Area Risk Classification

Legal uncertainty has long been recognized as a central and unavoidable feature of tax law, rather than a drafting defect or pathological deviation. Within legal theory, the principle of certainty of taxation is typically framed as a normative ideal—one that demands clarity, predictability, and stability in the definition and application of tax obligations—yet one that can never be fully realized in practice. As Demin argues, uncertainty in tax law operates along a continuum and manifests both negatively, as legislative omissions or inconsistencies, and positively, as a deliberate regulatory technique that enables flexibility, adaptability, and case-specific judgment in complex and evolving economic environments [9].

Our approach operationalizes specific manifestations of legal uncertainty in ways that are computationally tractable and empirically observable. Through discussions with tax law experts, reconstruction of avoidance schemes, and a review of the legal uncertainty literature, we identify a set of features that are tangible, precise, and amenable to explicit decision rules. These *gray area triggers* (Table 1) signal the potential for exploitative or abusive use of legal provisions. While the presence of a trigger does not constitute avoidance per se, it allows us to systematically identify risk-prone features embedded in tax legislation.

4.2 Expert-Informed Annotation

The corpus comprises all legal norms governing Colombia's Free Trade Zone (FTZ) regime from 1958 to 2022. This legislative domain was selected because it has been repeatedly identified as a focal point for tax avoidance strategies and exhibits frequent regulatory reforms that often close existing loopholes while simultaneously creating new ones. Each article was annotated for the presence of gray area triggers following predefined annotation guidelines and decision rules. Relevance, completeness, discretionality, and differential regime were treated as binary features (present or absent). Interpretability was annotated using an ordinal scale reflecting increasing degrees of interpretive openness: low (1), moderate (2), and high (3).

4.3 Embedding-Based Analysis

We examine how expert annotations are encoded in semantic space to assess whether gray area triggers correspond to systematic patterns in language representations. Specifically, we ask whether triggers form distinguishable clusters, whether particular triggers are associated with specific regions of embedding space, and whether embeddings reveal overlaps or interactions not immediately apparent from the annotation scheme itself.

Table 1: Definitions of gray area triggers.

Trigger	Type	Definition	Rationale
Relevance	Binary	An article contains a relevant fragment if it directly affects tax liabilities or regulates economic operations that have a direct impact on the calculation or payment of taxes.	Focusing on tax-relevant provisions isolates norms that can materially alter tax outcomes and therefore constitute potential entry points for strategic behavior.
Interpretability	Multiclass	An article contains an interpretable fragment if its wording admits multiple plausible interpretations in common or technical usage, potentially leading to different legal or tax consequences.	Interpretive flexibility allows alternative readings of the same provision to produce different tax treatments, generating gray areas rooted in semantic ambiguity.
Completeness	Binary	An article contains an incomplete fragment if it specifies legal or tax treatment for a subset of cases without exhaustively covering all relevant situations.	Partial specification leaves unregulated margins where analogous cases arise, creating uncertainty in the application of the norm.
Differential Regime	Binary	An article contains a differential regime fragment if it assigns different tax treatments to agents within the same system based on specific attributes or classifications.	Differential treatment creates incentives to adjust attributes or organizational forms in order to access preferential tax regimes.
Discretionality	Binary	An article contains a discretionary fragment if the application or enforcement of tax rules depends on administrative judgment rather than automatic legal criteria.	Discretion leads to heterogeneous enforcement and uncertainty, especially under limited auditing capacity, enabling gray areas in practical compliance.

4.4 Classification Task

We evaluate whether language models can learn expert-informed assessments of gray-area risk. We compare multiple text representation strategies, including TF–IDF as a baseline, Legal-BERT on its light spanish variant [5], and MEL [22], a benchmark model trained on Spanish legal texts. Logistic regression is used as the classification model across all configurations, which provides an interpretable and widely adopted baseline in legal NLP applications [7, 6].

4.5 Expert Annotation versus Language-Based Signals

Two complementary hypotheses guide the analysis. First, expert annotations are assumed to encode accumulated knowledge derived from legal practice and prior exposure to avoidance strategies, capturing latent regularities that are not fully explicit in the legal text. The classification task evaluates whether language-based representations can internalize and generalize this expert-informed signal. Second, expert criteria may not exhaust all potential sources of gray areas. Additional exploitability may arise directly from the layers of language itself, including lexical choice, syntactic structure, and semantic composition. We therefore examine how gray area trigger profiles change when the analysis relies solely on language-based representations without expert labels, and assess which dimensions of risk remain detectable under this restriction.

4.6 Robustness and Sensitivity Analysis

For the best-performing model configurations, we conduct robustness and sensitivity analyses to distinguish reliance on surface-level cues from more stable semantic representations. Following [21] and [23], and to see whether gray area detection is driven primarily by lexical patterns, syntactic configurations, or abstract semantic representations, we apply label-preserving perturbations to the input texts, including random reordering of phrases to test syntactic sensitivity and synonym replacement to test lexical robustness. Models that rely on genuine semantic signals are expected to maintain consistent predictions under semantically equivalent but formally altered inputs.

4.7 Legal Complexity and Predictive Performance

Prior literature documents a systematic relationship between tax law complexity and tax avoidance, though the underlying mechanisms remain contested. Complexity may overwhelm less sophisticated taxpayers while simultaneously creating opportunities for sophisticated actors to exploit ambiguities and structural gaps. We examine how the complexity of each legal article relates to predictive performance in gray-area classification. Two opposing mechanisms are theoretically plausible. On the one hand, increasing complexity—manifested in longer sentences, nested clauses, extensive enumerations, and dense cross-references—may introduce noise and hinder model performance. On the other hand, features associated with legal uncertainty and gray areas, such as incompleteness, conditional structures, and discretionary procedures, may be inherently intertwined with complexity. In this case, complexity may enhance predictive performance as a carrier of both structural and interpretive signals. We assess which mechanism dominates by relating article-level complexity measures to classification outcomes.

5 Results

5.1 The patterns on the expert-annotated data

The expert-annotated categories exhibit a clear class imbalance, with a systematic overrepresentation of label 0, corresponding to the absence of the trigger. This imbalance motivates the use of macro-averaged evaluation metrics, as they weight all classes equally and prevent overall performance measures from being dominated by the majority class. Figure 1 reports the distribution of articles by category and label, illustrating that the positive classes are comparatively sparse across all dimensions.

Figure 2 presents the UMAP projections of the MEL embeddings for each category. Interpretability and discretionality do not display clear clustering or separation in the embedding space, suggesting that articles annotated as positive for these dimensions are not distinguished by consistent lexical or semantic

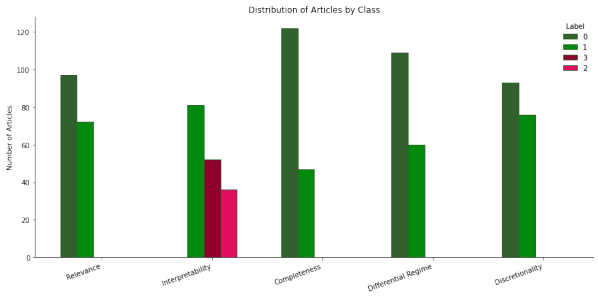
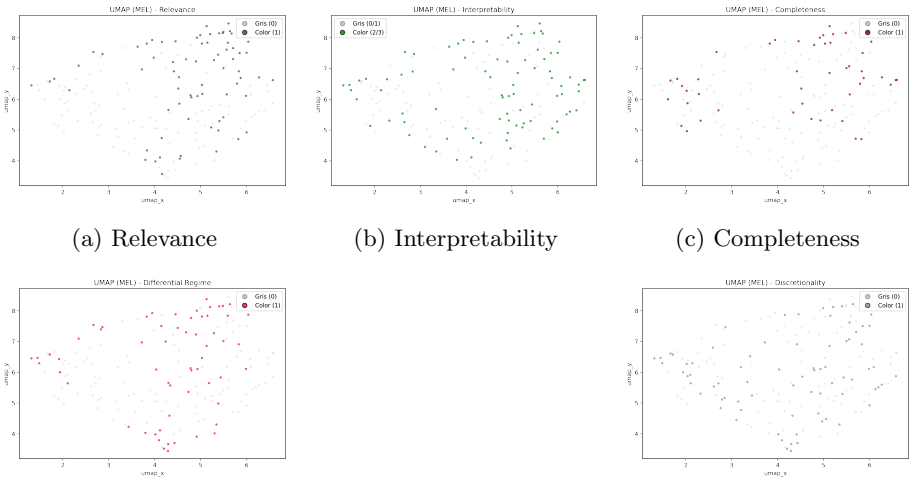


Figure 1: Distribution of article by gray-area trigger

patterns captured by the embeddings. This indicates either a broad or heterogeneous definition of these categories, or the presence of expert criteria that are not directly reflected in surface-level linguistic similarity. In contrast, relevance, completeness, and differential regime exhibit partial grouping and directional shifts in the embedding space, indicating a higher degree of alignment between expert annotations and latent semantic structure.



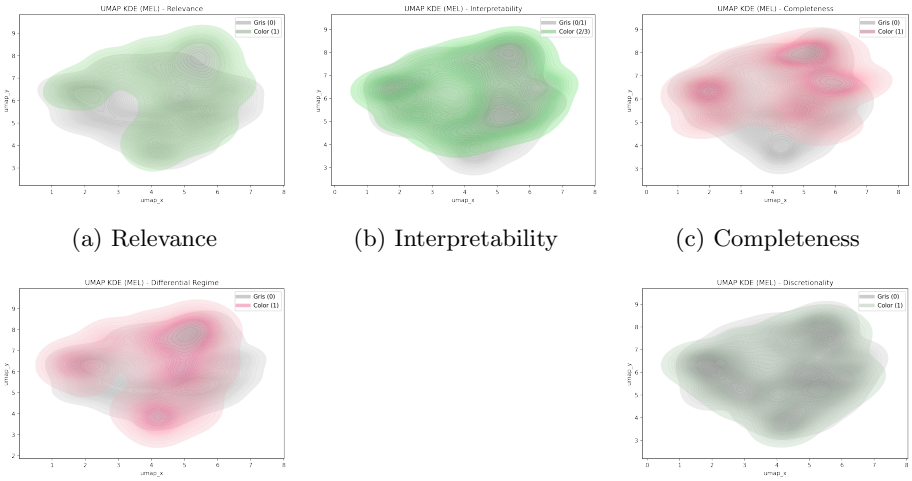
(a) Relevance (b) Interpretability (c) Completeness

(d) Differential Regime (e) Discretionality

Figure 2: Scatter plots of embeddings using MEL and UMAP.

Figure 3 presents the kernel density estimates for articles with and without the trigger in the embedding space. For interpretability and discretionality, the density distributions largely overlap, with no clear separation between labels.

For relevance, completeness, and differential regime, the distributions show small but consistent shifts, with localized regions of higher density for positive cases, despite substantial overlap overall.



(a) Relevance (b) Interpretability (c) Completeness

(d) Differential Regime (e) Discretionality

Figure 3: KDE plots with MEL embedding and UMAP by trigger.

Figures 4 and 5 compare articles with at least one risk category and articles with higher aggregated risk scores to low-risk articles in the embedding space. Both the scatter plots and the KDE visualizations display substantial overlap across groups, with no distinct clustering or density separation for high-risk provisions.

Since the risk score is defined as the simple sum of category-level values, the observed overlap indicates that aggregated risk is not reflected in global linguistic similarity as captured by the embeddings. The lack of separation does not negate the validity of expert annotations, but indicates that the criteria underlying risk assessment are not encoded in the textual representations employed in this analysis.

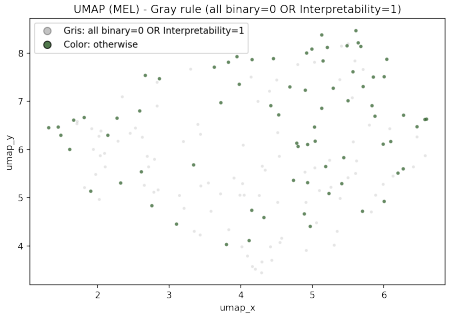


Figure 4: Scatter plot of the high risk articles against the low risk articles.

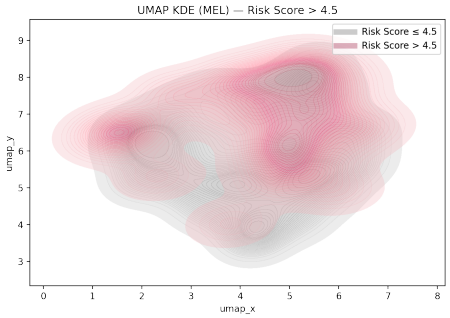


Figure 5: KDE plot of the high risk articles against the low risk articles.

5.2 Classification Task

Table 2 reports the comparative performance of TF–IDF, LegalELECTRA, and IIC/MEL across categories. TF–IDF achieves performance levels close to those obtained with MEL embeddings in most tasks, while LegalELECTRA performs consistently poorly and does not exceed a random baseline, particularly in terms of F1-score. Probably, that transformer-based model did not effectively learn the classification task under the current setup.

The classification signal is largely driven by surface-level textual features instead of deeper semantic representations. Category-level performance varies substantially, with differential regime yielding the highest scores across models and interpretability exhibiting the weakest performance, consistent with the lack of structure observed in the embedding analyses.

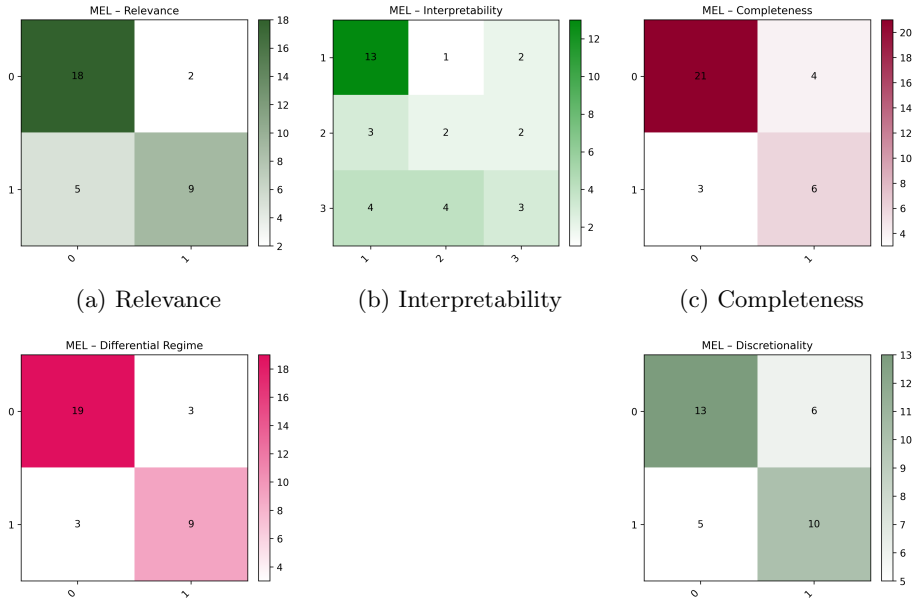
Table 2: Model performance comparison by category

	TF–IDF				LegalELECTRA				IIC/MEL			
Target	Acc	Prec	Rec	F1	Acc	Prec	Rec	F1	Acc	Prec	Rec	F1
Completeness	0.824	0.814	0.702	0.730	0.265	0.132	0.500	0.209	0.794	0.738	0.753	0.744
Differential Regime	0.882	0.868	0.890	0.875	0.353	0.176	0.500	0.261	0.824	0.807	0.807	0.807
Discretionality	0.765	0.766	0.754	0.757	0.559	0.279	0.500	0.358	0.676	0.674	0.675	0.674
Interpretability	0.471	0.262	0.343	0.272	0.471	0.157	0.333	0.213	0.529	0.455	0.457	0.447
Relevance	0.765	0.757	0.757	0.757	0.412	0.206	0.500	0.292	0.794	0.800	0.771	0.779

Note: identical data were used for all three models: $n = 169$ articles, with a train–test split of $n_{\text{train}} = 135$ and $n_{\text{test}} = 34$, using *Spanish text* as input. All metrics are macro-averaged. *Interpretability* is a multiclass task (classes 1, 2, 3); all other tasks are binary (classes 0, 1).

For the binary tasks, relevance, completeness, differential regime, and discretionality show a predominance of true negatives, reflecting the underlying class imbalance. Differential regime exhibits the most balanced distribution of true positives and true negatives, consistent with its higher overall performance.

In contrast, interpretability, modeled as a multiclass task, displays substantial confusion across classes, with frequent misclassification between adjacent labels.



(a) Relevance (b) Interpretability (c) Completeness

(d) Differential Regime (e) Discretionality

Figure 6: Confusion matrices for MEL across the five targets.

5.3 Expert Annotation versus Language-Based Signals

Struggling with the code, but a hair far from there

5.4 Robustness and Sensitivity Analysis

Struggling with the code, but a hair far from there

5.5 Legal Complexity and Predictive Performance

Consistent with prior findings in the literature, expert annotations under the gray-area framework exhibit an association with the structural and textual complexity of legal provisions. Figure 7 shows that articles with lower complexity levels tend to be associated with lower risk scores, while articles with higher complexity are more frequently linked to higher risk scores.

The plot partitions the space using the midpoints of the complexity index and the risk score, yielding four quadrants that facilitate a descriptive comparison between low- and high-complexity provisions and their corresponding risk levels.

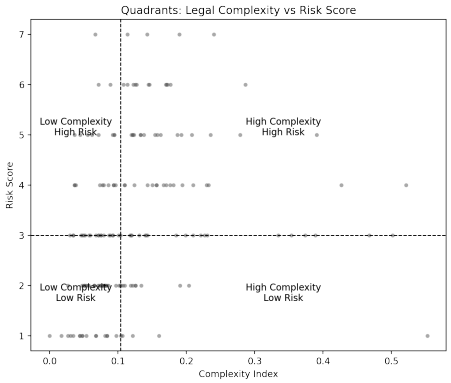


Figure 7: Legal Complexity vs Risk Score

Figure 8 presents the distribution of risk scores across different levels of legal complexity. The results show that higher complexity levels are associated with higher median risk scores. In addition, the dispersion of the risk score increases with complexity, which indicates greater variability in uncertainty among more complex legal provisions.

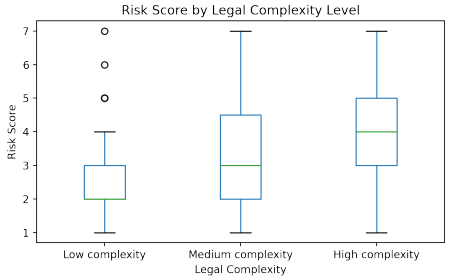


Figure 8: Risk Score by Legal Complexity Level

Figure 9 shows the distribution of the complexity index across categories. Notably, the categories with weaker classification performance, namely interpretability and discretionality, exhibit a wider dispersion in complexity values. In contrast, categories with stronger classification results display comparatively lower variability in complexity, particularly for relevance, and to a lesser extent for differential regime and completeness.

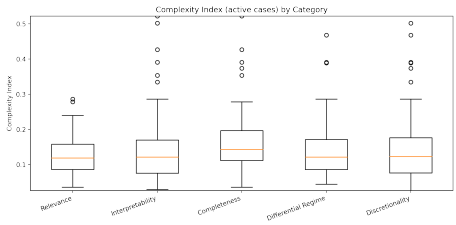


Figure 9: Complexity index distribution by category

6 Discussion

As noted above, this study examines gray-area triggers linked to expert-identified sources of tax norm abuse as an initial step toward anticipating potential exploitative uses directly from statutory language. The scope of the analysis is intentionally restricted to uncertainty-based features and does not incorporate other mechanisms emphasized in prior research, such as interactions across legal provisions [4].

Within these bounds, the primary contribution of the paper is the construction of the first annotated dataset of tax law provisions explicitly oriented toward norm abuse. The empirical results indicate that certain gray-area categories correspond to identifiable textual regularities, while others do not. This differentiation offers a concrete basis for subsequent work, either through the refinement of categories that remain overly broad or through the incorporation of non-textual information, including legal interactions and enforcement signals, into computational approaches to tax avoidance detection.

The findings show that models operating on legal text can anticipate gray-area triggers that rest on explicit linguistic structure, including differential treatment, scope delimitation, and normative completeness. In these cases, the statutory language itself provides sufficient signal for systematic identification.

In contrast, gray areas associated with interpretability and discretionality do not appear to be predictable from the written norm alone. These categories lack stable textual structure within the embedding space, which suggests that their exploitability depends on contextual, institutional, or strategic factors that remain external to the statutory formulation.

These ideas support the literature that suggest the integration of uncertainty-based textual approaches with methods that account for cross-provision interactions and enforcement dynamics in future research.

7 Conclusion

We conclude that a model trained on legal-domain Spanish texts can detect gray-area triggers in tax legislation with high effectiveness, achieving accuracy

levels close to 80%. For several categories, textual information provides sufficient signal to support reliable classification.

The strongest results correspond to differential regime, relevance, and completeness. These categories display structured organization in the embedding space and consistent predictive behavior. By contrast, interpretability and discretionality show limited predictive performance and lack clear structure in the vector representations. This pattern indicates that the former categories align with latent linguistic regularities, while the latter do not rely on stable textual features.

Accordingly, expert assessments of interpretability and discretionality do not appear to derive from properties encoded in the statutory text itself. This outcome motivates further analysis to determine whether these categories remain overly broad or whether expert judgment draws on dimensions that fall outside textual representation.

The results also reveal an association between legal text complexity, the presence of gray-area triggers, and model performance. More tractable textual structures facilitate the identification of relevant features, even when the task relates closely to legal complexity. Future computational approaches to the identification and exploration of gray areas should therefore treat textual complexity as an explicit dimension, in order to reduce biased or confounded detection outcomes.

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